

BT3041 – Analysis and Interpretation of Biological Data

Assignment 3

Instructions

Please provide your analysis results in a report format, specifically answering each question with relevant plots, etc. Also, state any assumptions made clearly in the report. Attach the Google Drive link to your software code (Python) used to perform the calculations for the report.

Submit your completed assignment report and supplementary files (if any) in Turnitin (Link for submission will be shared separately) by 17/04/2026 23:59:00. Any late submissions beyond this timeline will incur a penalty of 5 marks per hour. Obviously, if you submit it later than 10 hours, the entire assignment is considered forfeited. Plagiarism and the use of AI are strictly discouraged.

Questions:

Total marks: 50

Download the [wine dataset](#). The data contains 11 features describing wine characteristics, and the 'target' column indicates the wine class. This target is the ground truth.

1. Do agglomerative clustering using Ward linkage, cut at 3 clusters, and plot a heatmap with a dendrogram. Label this dendrogram based on the true labels, calculate the ARI & AMI, and report your observations. [10]
2. Cluster the data using single linkage and complete linkage with Euclidean distance and Manhattan distance separately, and compare their performance using ARI, AMI, and reason which linkage and distance metric you think is better. [10]
3. Find the optimal number of k-means clustering using the elbow plot of the silhouette score and Calinski-Harabasz index. [10]
4. Perform k-means clustering with the optimal number of clusters found in (3) and evaluate the clustering using ARI and AMI. Infer your observations. [10]
5. Use DBSCAN to estimate the number of outliers with epsilon 2.2, perform hierarchical clustering and k-means clustering with the outliers removed, and compare the performance with (1) and (4). [10]

Hint: 'target' needs to be used only when evaluating or labeling, not for the analysis. Consider using the rule of thumb for choosing the minimum number of points ($D+1$) for DBSCAN.